



# UNIVERSITY OF THE PUNJAB

BS (After Associate Degree) : Fifth Semester – Fall 2023

Paper: Computational Physics – I

Course Code: PHYS-3502

Time

60

**THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED**

Q.1. Solve the following questions.

(6×5=30)

- Write C++ program print v and s such that  $v=u+at$  and  $s=ut+\frac{1}{2}at^2$  take input from user for three iterations.
- Explain with examples a) for loop b) int and c) pow()
- Write program to calculate A and V such that  $A=4\pi r^2$  and volume  $V=\frac{4}{3}\pi r^3$  Take radius values from the user. Implement with arrays.
- Store and print 9 random numbers into a matrix A (3×3). Also print A values.. Find and print transpose of A values.
- Write program to print x vs y such that  $y(x) = x^4+3x^3-7x-28$  Take x values from the user for 10 iterations.
- Write program to calculate and print force value by taking inputs for x and k values from the user for 10 iterations.

Solve the following questions.

(3×10=30)

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Q.2.) | Write C++ program to evaluate the integral $\int_1^6 \left(\frac{\sqrt{x}+1}{3}\right) dx$ by both trapezoidal rule and Weddle's rule (with n=6)<br>⇒ Write C++ program to generate twenty one random-number pair of points (X,Y) and calculate (20) distance values of consecutive points. Discuss how these points show Brownian motion?                                                                                                                                                                                                                                                                                                                                                     | 6+4 |
| Q.3.  | Suppose G and H be 3×3 matrices. Write C++ program which reads in entries of the matrices and calculate (i) $J=2G+H$ , (ii) $K=G \times H$ (iii) $L=G-H$ (iv) transpose of J (v) mean of values of G.<br>⇒ Describe the use of an array. Write program to calculate $c=a^2-b^2$ and $d=a+b$ . Implement your program using loops and arrays.                                                                                                                                                                                                                                                                                                                                                   | 6+4 |
| Q.4.) | Write C++ program for the Damped harmonic motion (D.H.M) of a mass attached with a spring using Euler's method under the following conditions: ( $g=9.8 \text{ m/s}^2$ , initial position zero and velocity 15 m/s, time step 0.1 sec. and maximum time 15 sec., $k=1 \text{ N/m}$ , $m=1 \text{ kg}$ , damping coefficient = 0.5 N/ms, $\gamma=0.01 \text{ s}^{-1}$ and $f_0=1.5 \text{ N}$ ). Calculate and print values for time, position, velocity and acceleration. How you can change the same program for Simple H.M., and Forced H.M.<br><br>Write C++ program which reads in a number as decimal number and convert it into binary number. Implement your program for 10 iterations. | 5+5 |



# Computational Physics-I

5<sup>th</sup> semester fall 2023.

Question #01:

Q1

```
#include <iostream.h>
#include <conio.h>
void main()
{
    clrscr();
    float u, a, v, t, s;
    for (int i=0; i<3; i++)
    {
        cout << "Enter the value  
of u for iteration << i+1 << " : ";
        cin >> u;
        cout << "Enter the value of a  
for iteration << i+1 << " : ";
        cin >> a;
        cout << "Enter the value of  
t for iteration << i+1 << " : ";
        cin >> t;
```



```

v = u + a * t;
s = a * t + 0.5 * a * t * t;
cout << "v=" << endl;
cout << "s=" << endl;
}
getch();
}

```

Ex 2

### (a) For loop

A for loop is a type of loop that allows you to execute a block of code repeatedly for specific number of iteration.

Syntax:

→ (Initialization; condition; Increment/Decrement)

e.g:

```

for (int i=0; i <= 7; i++)
{
    cout << i << endl;
}

```

This will output:

1

2

3

4

5

6

(b) Int:

Size of int is 2 bytes  
(1024 bits makes 1 byte). Int in  
C++ represents an integer value.

It is a whole number,  
either positive, negative or zero,  
without a fractional part.

e.g:

int x = 6;

cout << x << endl;

This will output:

6

(c) Pow():

Pow() is the mathematical  
function in C++ that calculate  
the values of a number  
raised to a power.



It takes two arguments:

- base
- exponent.

e.g.:

```
double result = pow(2, 3);  
cout << result << endl;
```

This will output:

8

~~ex iii~~

$\therefore r = \text{numRadii}$

```
#include <iostream.h>
```

```
#include <conio.h>
```

```
#include <math.h>
```

```
void main()
```

```
{ // Number of radii values
```

```
int numRadii = 5;
```

```
// array of radii values
```

```
float [numRadii] = {1.0, 2.0, 3.0, 4.0, 5.0};
```

```
// Surface area and volume
```

```
float A, V;
```

```
for (int i = 0; i <= numRadii; i++)
```

```
{ // calculate surface area
```

```
A = 4 * pi * r.2;
```

```
// calculate volume
```

```
V = (4.0 / 3.0) * pi * r.3;
```



```

cout << "Radius:" << radii[i] << endl;
cout << "Surface area:" << A << endl;
cout << "Volume:" << V << endl;
}
getch();
}

```

Q.10

Void main()

```

{ srand(time(0));
  int A[3][3];
  for (int i=0; i<3; i++)
  {
    for (int j=0; j<3; j++)
    {
      A[i][j] = rand()%10;
      cout << A[i][j] << endl;
    }
  }
  cout << "Transpose of A =" << endl;
  for (int i=0; i<3; i++)
  {
    for (int j=0; j<3; j++)
    {

```

```
cout << A[i][i] << " ";
```

```
}
```

```
cout << endl;
```

```
}
```

```
getch();
```

```
}
```

— ~~cevi~~ —

```
Void main ()
```

```
{
```

```
for(int i=0; i<=10; i++)
```

```
double x, y;
```

Enter the value of x

```
for iteration << i+1 << endl;
```

```
cin >> x;
```

```
y = pow(x, 4) + 3 * pow(x, 3)
```

```
    - 7 * x - 28;
```

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```
cout << "x = " << x << "y = "
```

```
<< y << endl;
```

```
}
```

```
getch();
```

```
}
```

— ~~cevi~~ —



```

void main()
{
    for (int i=0; i<10; i++)
    {
        float x, k, F;
        cout << "Enter the value
        of x for iteration << i+1 <<
        endl;
        cin >> x;
        cout << "Enter the value of k
        for iteration << i+1 << endl;
        cin >> k;
        F = k * x;
        cout << "Force value for iteration"
        << i+1 << F << endl;
        getch();
    }
}

```

Question #02

(a)

```

float f (float b)
{
    b = sqrt(x) + 1.0 / 3.0;
    return b;
}

```



```
void main()
```

```
{ cout << "Trapezoidal rule" << endl;
```

```
float h, n, up, low, sum;
```

```
cout << "Enter the upper limit;
```

```
cin >> up;
```

```
cout << "Enter the lower limit";
```

```
cin >> low;
```

```
n = (b - a) / h;
```

```
h = (b - a) / n;
```

```
sum = f(low) + f(up);
```

```
for (i = 1; i < n; i++)
```

```
{ sum += f(low + i * h);
```

```
sum = sum * h / 2;
```

```
cout << "Trapezoidal rule" << sum << endl;
```

```
} cout << "Needles rule" << endl;
```

```
for (i = 1; i < n; i++)
```

```
{ PUACP
```

```
if (i % 2 == 0)
```

```
{ sum += 2 * f(low + i * h);
```

```
}
```

```
else
```



```

{
    Sum += 4 * f(low + i * h);
}
Sum = Sum * h / 3;
}
cout << "Weddles sum" <<
    "I =" << sum << endl;
getch();
}

```

void main()

```

{
    srand(time(0));
    for(int i=0; i<21; i++)
        // Generate 21 random No.
        float x = (rand() % 10);
        float y = (rand() % 10);
        cout << "Point" << i+1 << " : (" << x
        << " , " << y << ")" << endl;
    }
}

```

// Calculate Euclidean distance  
b/w constructive points.

```

for (int i=0; i<20; i++)

```



```

float x1 = (rand() * 10);
float y1 = (rand() * 10);
float distance = sqrt((x2 - x1)
    * (x2 - x1) + (y2 - y1) * (y2 - y1));
cout << "Distance b/w two points"
<< i+1 << " and " << i+2 << ":"
<< distance << endl;
}
getch();
}

```

Question #03: a)3

(i)  $\rightarrow J = 2G + H$

```

for (int i = 0; i < 3; i++)
{
    for (int j = 0; j < 3; j++)
    {
        J[i][j] = 2 * G[i][j] +
            H[i][j];
    }
}

```

(ii)  $\rightarrow K = G - H$

```

for (int i = 0; i < 3; i++)
{

```



```

for (int j = 0; j < 3; j++)
{
    L[i][j] = G[i][j] - H[i][j];
}
}

```

(iv) → Transpose of J.

```

for (i = 0; i < 3; i++)
{
    for (j = 0; j < 3; j++)
    {
        transposeJ[j][i] = matrix[i][j];
    }
}
cout << "Transpose of matrix J"
<< endl;

```

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(v) →  $K = G \times H$

```

for (int i = 0; i < 3; i++)
{
    for (int j = 0; j < 3; j++)
    {
        K[i][j] = 0;
        for (int k = 0; k < 3; k++)
        {
            K[i][j] += G[i][k] * H[k][j];
        }
    }
}

```



(V) Mean of value G

```
{
    double sum = 0;
    for (int i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
            sum += G[i][j];
        }
    }
    Mean = sum / 9;
}
```

On top to declare 3x3 matrix:

```
#include <iostream.h>
```

```
#include <conio.h>
```

```
using namespace std;
```

```
{ clrscr();
```

```
for (int i = 0; i < 3; i++)
```

```
{
```

```
for (int j = 0; j < 3; j++)
```

```
{
```

```
cout << matrix[i][j] << endl;
```

```
}
```

```
}
```



for output:

```
cout << "Matrix J:" << endl;
cout << "Matrix K:" << endl;
cout << "Matrix L:" << endl;
cout << "Transpose of J:" << endl;
cout << "Mean of values in G:"
    << G << endl;
    getch();
}
```

array

Array:

An array is a data structure that stores a collection of elements, each identified by an index or key. In programming, arrays are used to store and manipulate large amount of data efficiently.

```
void main()
{
```

```
    int size = 5;
```

```
    int a[size] = {1, 2, 3, 4, 5};
```

```
    int b[size] = {6, 7, 8, 9, 10};
```

```
    int c[size];
```



```

int d[size];
for (int i=0; i < size; i++)
{
    c[i] = a[i] * a[i] - b[i] *
        b[i];
    d[i] = a[i] + b[i];
}
for (int i=0; i < size; i++)
{
    cout << "a=" << a[i] <<
        ", b=" << b[i] << endl;
    cout << "c = a2 - b2 =" <<
        c[i] << endl;
    cout << "d = a+b" << d[i] <<
        endl;
}
getch();
}

```

**PUACP**  
**Question #04:**  $(a = -\frac{k}{m}x - \frac{c}{m}v)$

(i) **Damped Harmonic Motion.**

float g = 9.8;

float k = 1, m = 1, c = 0.5;

float sh = 0.1, Lmax = 15, fmax = 7.5;



```
float t=0, x=0, v=15, a=0, F=0;
float il=1;
```

```
while (t <= tmax)
```

```
{ ta(il) = 0;
```

```
xa(il) = x;
```

```
va(il) = v;
```

```
aa(il) = a;
```

```
a = (-(k*x) - (c*v)) / m;
```

```
x = x + v*h;
```

```
v = v + a*h;
```

```
t = t + h;
```

```
il = il + 1;
```

```
cout << "t" << x << "v" << a << endl;
```

```
{
```

```
}
```

في وقت لاحق

```
}
```

→ Convert this for SHM

```
while (t <= tmax)
```

```
{ ta(il) = t;
```

```
xa(il) = x;
```

```
va(il) = v;
```

```
a = -(k*x) / m;
```



```

int d[size];
for (int i=0; i < size; i++)
{
    c[i] = a[i] * a[i] - b[i] *
        b[i];
    d[i] = a[i] + b[i];
}
for (int i=0; i < size; i++)
{
    cout << "a=" << a[i] <<
        "b=" << b[i] << endl;
    cout << "c = a2 - b2 =" <<
        c[i] << endl;
    cout << "d = a+b" << d[i] <<
        endl;
}
getch();
}

```

**PUACP**  
**Question #04:**  $(a = -\frac{k}{m}x - \frac{c}{m}v)$

(i) Damped Harmonic Motion.

```
float g = 9.8;
```

```
float k = 1, m = 1, c = 0.5;
```

```
float xh = 0.1, tmax = 15, fmax = 1.5;
```



```

float t=0, x=0, v=15, a=0, F=0;
float il=1;
while (t <= tmax)
{
    ta(il) = a;
    xa(il) = x;
    va(il) = v;
    aa(il) = 0;
    a = (-(k*x) - (c*v))/m;
    x = x + v*h;
    v = v + a*h;
    t = t + h;
    il = il + 1;
    cout << t << "t" << x << "x" << v << "v" << a << endl;
    getch();
}

```

→ Convert this for SHM

```

while (t <= tmax)
{
    ta(il) = t;
    xa(il) = x;
    va(il) = v;
    a = -(k*x)/m;
}

```



$$aa(il) = a;$$

$$x = x + v * h;$$

$$v = v + a * h;$$

$$t = t + h;$$

$$il = il + 1;$$

}

→ Convert this into FHM

$$a = \frac{-k}{m} x - \frac{c}{m} v + \frac{F_0}{m} \cos \omega t$$

while (t < t<sub>max</sub>)

{ aa(il) = a;

ta(il) = t;

xa(il) = x;

va(il) = v;

t = t + h;

il = il + 1;

a = (-k \* x) / m - (c \* v) / m + (F<sub>0</sub> \*  
cos(ω \* t)) / m;

x = x + v \* h;

v = v + a \* h;

}



c++

```
void main()
```

```
{ int decimal;
```

```
for (int i = 0; i < 10; i++)
```

```
{
```

```
cout << "Enter a decimal  
number: ";
```

```
cin >> decimal;
```

```
cout << "Binary representation: ";
```

```
for (int j = 31; j >= 0; j--)
```

```
{
```

```
cout << ((decimal >> j) & 1) << endl;
```

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```
}
```

```
getch();
```

```
PUACP
```